Comparative Analysis – Network Topologies

Question 1: Compare Star Topology and Bus Topology in terms of Cost, Reliability and Performance.

Criteria	**Star Topology**	**Bus Topology**
Cost	Generally more expensive because it requires a central switch or hub and separate cables for each node.	Less expensive because it uses a single backbone cable.
Reliability	More reliable because failure of one node does not affect the other nodes.	Less reliable because failure of the main backbone cable can bring down the entire network.
Performance	Better performance because data transmission is managed through a central device, reducing collisions.	Performance decreases as more devices are added because of increased traffic on the shared medium.
Conclusion	Better in reliability and performance.	Better when low cost and simple installation are priorities.

Question 2: Compare Mesh Topology and Star Topology in terms of Fault Tolerance.

Criteria	**Mesh Topology**	**Star Topology**
Fault Tolerance	Very high fault tolerance because multiple paths are available between devices.	Moderate fault tolerance because individual node failures do not affect others.
Link Failure	If one link fails, data can use an alternate path.	Failure of a connection to a node affects that particular node.
Central Device Failure	No single central device controls the entire network.	Failure of the central hub or switch can stop the entire network.
Conclusion	More robust and reliable for critical environments.	Less fault tolerant than mesh but easier to manage.

Question 3: Compare Bus Topology and Mesh Topology in terms of Installation Complexity.

Criteria	Bus Topology	Mesh Topology
Structure	Uses a single backbone cable with minimal connections.	Each node is connected to multiple or all other nodes.
Installation	Simple and easy to install.	Complex and time-consuming to install.
Cabling	Requires relatively less cabling.	Requires a large number of cables and ports.
Scalability Impact	Simple for small networks but becomes less suitable as devices increase.	Complexity increases significantly as more devices are added.
Conclusion	Much easier and cheaper to install.	More complex and costly to install.

Question 4: Compare Hybrid Topology and Star Topology in terms of Scalability.

Criteria	Hybrid Topology	Star Topology
Scalability	Highly scalable because different topologies can be combined and new segments can be added flexibly.	Scalable to a certain extent, depending on the capacity of the central switch or hub.
Expansion	New network segments can be added without significantly affecting existing segments.	Additional switches or hubs may be needed when the central device reaches its port limit.
Flexibility	Very flexible because it can combine star, bus, mesh and other topologies.	Less flexible because it mainly depends on a central device.
Conclusion	Provides greater scalability and flexibility.	Provides good scalability but has limitations based on central-device capacity.

Question 5: Compare Star, Mesh, Bus and Hybrid Topologies in terms of Maintenance.

Topology	Maintenance Characteristics	Overall Assessment
Star	Faults can be quickly identified and isolated to a specific node or cable. Centralized management makes troubleshooting easier.	Easy to maintain.
Mesh	Large number of connections makes troubleshooting and maintenance time-consuming and complex.	Difficult to maintain.
Bus	Backbone cable faults can affect the entire network and can be difficult to locate.	More difficult to maintain than star.
Hybrid	Combines multiple topologies, so maintenance requires careful monitoring and skilled administration.	Moderate maintenance complexity.
Conclusion	Star is easiest to maintain; mesh is the most complex; bus is difficult because of backbone faults; hybrid has moderate complexity.	Star is generally the easiest for maintenance.